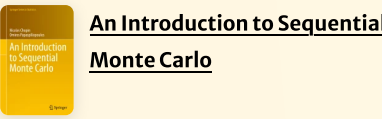


Importance Resampling

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Summary

Resampling is the action of drawing randomly from a weighted sample, so as to obtain an unweighted sample. Resampling may be viewed as a random weight importance sampling technique. However, it deserves a separate chapter because it plays a central role in particle filtering. In particular, we explain that resampling has the curious property of potentially reducing the variance at a later stage, provided that this later stage corresponds to a Markov update that forgets its past in some way. This point is crucial for the good performance of particle algorithms.

This chapter explains informally this particular property, formalises importance resampling, describes several algorithms to perform resampling, and compares these algorithms numerically.

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